Cover letter

14 October 2026

The Editor-in-Chief
Endocrine Connections

Dear Editor-in-Chief,

Please consider “TSHR and IGF1R in Graves disease and thyroid eye disease: genetic evidence from blood expression” as an original research article.

Teprotumumab blocks IGF-1R and is effective in thyroid eye disease (TED). That success raises the question of whether *IGF1R* also carries inherited susceptibility to Graves disease (GD). Therapeutic efficacy and disease susceptibility are different claims, and we asked how consistently genetic evidence links blood expression of *TSHR* and *IGF1R* to disease. Within a Mendelian randomization screen of 2,544 druggable genes, we used Biobank Japan GD for discovery and compared UK Biobank hyperthyroidism and three overlapping FinnGen endpoints: GD, Graves ophthalmopathy and strict Graves ophthalmopathy.

Higher genetically proxied *TSHR* expression was associated with lower GD odds in Biobank Japan ($P = 1.09\times10^{-14}$) and FinnGen GD ($P = 3.19\times10^{-32}$), in the same direction in all five datasets. Colocalization supported a shared causal variant in Biobank Japan and both ophthalmopathy endpoints (in Biobank Japan and the strict endpoint, not under the most conservative prior tested) and favored distinct nearby variants in UK Biobank and FinnGen GD, the two datasets with the most cases, pointing to candidate variants for fine-mapping. The protective direction fits central tolerance: the expression-increasing *TSHR* allele is coupled with the rs179247 allele previously linked to GD protection and greater thymic transcription. All *IGF1R* odds ratios exceeded 1, with nominal associations in two datasets; the most precise estimate, in FinnGen GD, was close to the null, compatible with at most a modest inherited effect.

Further analyses show where the evidence is robust: an allele-direction audit confirmed the *TSHR* direction, and allele-frequency substitution and instrument-omission analyses show where estimates depend on single variants. A fixed five-tissue GTEx extension and a small orbital RNA-sequencing series (four TED samples, one control) are reported as supplementary and descriptive; the adipose instruments come from the same *TSHR* region and are not presented as independent evidence.

The results also define clear next steps. Because the FinnGen endpoints overlap and use population controls, eye-disease susceptibility among people who already have GD is best addressed by comparing GD patients with and without TED. The findings concern susceptibility rather than treatment response and are compatible with the established benefit of IGF-1R blockade. The FinnGen extensions and tissue analyses were post hoc and are labeled as such.

The submission includes three main tables, three main figures, a supplement with four tables and two figures, a single results workbook and a STROBE-MR reporting crosswalk.

An earlier, narrower version was considered at thyroid-focused journals; the work was subsequently restructured around the broader genetic screen. The orbital dataset also underlies a separate whole-transcriptome analysis, which is not under review elsewhere; here only three genes are described, and no genome-wide expression or pathway catalogue is presented.

This manuscript has not been published and is not under consideration elsewhere. The study received no specific funding, and the authors report no competing interests. The orbital component was approved by the Pusan National University Hospital IRB (2104-018-102), with written informed consent.

Sincerely,

Suk-Woo Yang, MD, PhD
Corresponding author, on behalf of Jungyul Park, Min-Seon Kim and Suk-Woo Yang
Department of Ophthalmology, Seoul St. Mary’s Hospital
College of Medicine, The Catholic University of Korea, Seoul, Republic of Korea
E-mail: yswoph@catholic.ac.kr

Author institutions: J. Park and S.-W. Yang, Department of Ophthalmology, Seoul St. Mary’s Hospital, The Catholic University of Korea, Seoul; M.-S. Kim, Department of Ophthalmology, The Catholic University of Korea, Seoul; Republic of Korea.